

Commercial Pitch Report

Telugu Fake News & Misinformation Classifier

Project Overview

The Telugu Fake News & Misinformation Classifier is an AI-powered platform designed to detect fake, misleading, and unverified Telugu news shared through WhatsApp, social media, and digital platforms. Built using multilingual NLP and transformer-based architectures, the system classifies content into Real, Fake, or Unverifiable categories with explainable predictions. The model achieved an F1-score of 0.78, outperforming the GPT-4 baseline by 27.87%. It was trained on 41,318 Telugu-language samples collected from FACTIFY, IndicGLUE, RSS feeds, and verified regional news sources.

Problem Statement

Misinformation in regional Indian languages is growing rapidly, especially across WhatsApp and social media platforms. Most global fact-checking systems primarily support English and fail to understand Telugu language context, regional slang, cultural nuances, and local political narratives. This creates a major gap in digital trust and misinformation verification for Telugu-speaking communities.

Our Solution

The platform is built on Google's MuRIL (Multilingual Representations for Indian Languages) and fine-tuned specifically for Telugu fake news detection. The system provides real-time classification with confidence scores, LIME-based explainability highlighting trigger words, automatic English translation through Google Translate API, and keyword detection for suspicious patterns such as urgency prompts, scams, and manipulative messaging. The solution is deployed through a scalable Flask-based web application with a modern responsive interface.

Key Features

- Three-class classification: Real, Fake, Unverifiable
- Confidence scoring with probability distribution
- Explainable AI using LIME highlighting suspicious trigger words
- Real-time analysis in 5–10 seconds
- Scalable Flask web architecture
- Telugu input with English translation output
- Modern responsive user interface
- Suspicious keyword and scam pattern detection

Technical Stack

Backend: Flask 3.0, PyTorch 2.3.1, Transformers 4.44.2
Model: MuRIL (google/muril-base-cased)
Explainability: LIME
Translation: Google Translate API

Dataset: 41,318 Telugu samples from FACTIFY, IndicGLUE, RSS feeds, and verified news sources

Business Potential

The platform targets social media platforms, news organizations, fact-checking agencies, educational institutions, and government cybersecurity departments. With over 96 million Telugu speakers globally, the market opportunity for regional misinformation detection is significant. Revenue streams include API subscriptions, enterprise licensing, browser extensions, white-label deployments, and Telegram/WhatsApp bot integrations.

Revenue Streams

- API Subscription Model: Basic (\$99/month), Pro (\$499/month), Enterprise (custom pricing)
- White-label licensing for media and fact-checking organizations
- Browser extension with freemium and premium tiers
- WhatsApp and Telegram bot integrations with subscription plans

Go-to-Market Strategy

Phase 1: Partner with Telugu news outlets and launch browser extension to acquire early users.

Phase 2: Collaborate with fact-checking organizations and social media integrations.

Phase 3: Expand into other Indian languages including Tamil, Kannada, and Hindi while scaling enterprise adoption.

Competitive Advantage

The platform is one of the few Telugu-focused fake news detection systems with a proven 78% F1-score accuracy. It delivers real-time predictions within seconds, provides explainable AI insights, and includes multilingual translation support — features that many competitors currently lack.

Investment Opportunity

Seeking \$150,000 investment for a 12-month runway focused on multilingual expansion, cloud infrastructure scaling, enterprise partnerships, and product commercialization. Projected revenue targets include \$250K in Year 1, \$1.2M in Year 2, and \$4.5M in Year 3.

Conclusion

The Telugu Fake News & Misinformation Classifier delivers a production-ready AI solution for regional misinformation detection. Combining MuRIL, PyTorch, Flask, multilingual NLP, and explainable AI, the platform offers strong technical capabilities, scalable architecture, and high commercial potential for deployment across media, social platforms, and public-sector organizations.